Example 1.32

Is 400 a perfect cube?

Solution:

By prime factorisation, we have $400 = \underline{2 \times 2 \times 2} \times 2 \times 5 \times 5$.

There is only one triplet. To make further triplets, we will need two more 2's and one more 5. Therefore, 400 is not a perfect cube.

Example 1.33

Find the smallest number by which 675 must be multiplied to obtain a perfect cube.

Solution:

3	675
3	225
3	75
5	25
5	1
	1

We find that, $675 = 3 \times 3 \times 3 \times 5 \times 5$ .....(1)

Grouping the prime factors of 675 as triplets, we are left over with 5×5.

We need one more 5 to make it a perfect cube.

To make 675 a perfect cube, multiply both sides of (1) by 5.

$675 \times 5 = 3 \times 3 \times 3 \times 5 \times 5 \times 5$

$3375 = 3 \times 3 \times 3 \times 5 \times 5 \times 5$

Now, 3375 is a perfect cube. Thus, the smallest required number to multiply 675 such that the new number perfect cube is 5.



Think

In this question, if the word 'multiplied' is replaced by the word 'divided', how will the solution vary?

1.8.3 Cube root of a given number by Prime Factorisation

Step 1: Resolve the given number into the product of prime factors.

Step 2: Make triplet groups of same primes.

Step 3: Choosing one from each triplet, find the product of primes to get the cube root.

Example 1.34

Find the cube root of 27000.

Solution:

By prime factorisation, we have $27000 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 5 \times 5$

$$\therefore \sqrt[3]{27000} = 2 \times 3 \times 5 = 30.$$

Example 1.35

Evaluate: (i) $\sqrt[3]{\frac{9261}{8000}}$ (ii) $\sqrt[3]{\frac{1728}{729}}$

Solution:

(i) $\sqrt[3]{\frac{9261}{8000}} = \frac{\sqrt[3]{9261}}{\sqrt[3]{8000}} = \frac{(21 \times 21 \times 21)^{\frac{1}{3}}}{(20 \times 20 \times 20)^{\frac{1}{3}}} = \frac{\left(21^3\right)^{\frac{1}{3}}}{\left(20^3\right)^{\frac{1}{3}}} = \frac{21}{20} = 1\frac{1}{20}$

(ii) $\sqrt[3]{\frac{1728}{729}} = \frac{\sqrt[3]{1728}}{\sqrt[3]{729}} = \frac{(12 \times 12 \times 12)^{\frac{1}{3}}}{(9 \times 9 \times 9)^{\frac{1}{3}}} = \frac{\left(12^3\right)^{\frac{1}{3}}}{\left(9^3\right)^{\frac{1}{3}}} = \frac{12}{9} = \frac{4}{3} = 1\frac{1}{3}$

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